

StudyMate.AI

Vibe Coding: Building & Deploying an AI Web Application on AWS

Containerized Docker Application & AWS Elastic Beanstalk Report

LIVE AWS ELASTIC BEANSTALK URL:

<http://Studymate-AI-env.eba-6yuyfhhu.ap-southeast-2.elasticbeanstalk.com>

Submitted for: Vibe Coding Masterclass Series

Organization: IBM & Bharat Cares

PART 1: PROJECT CONCEPT NOTE

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1. Project Title & Application Name

- Application Name: StudyMate.AI
- Project Title: Containerized AI Learning Assistant on AWS Elastic Beanstalk

2. Problem Statement & Objective

Students and researchers face cognitive overload when synthesizing dense study material. StudyMate.AI simplifies notes into ELI5 summaries, answers real-time queries via an AI tutor, and auto-generates MCQ practice quizzes.

3. Target User & Use Cases

- High School & University Students preparing for exams.
- Researchers & Educators analyzing documents (.pdf, .docx, .txt).
- Self-Learners seeking interactive subject guidance.

4. LLM Model & API Integration

- LLM Engine: Google Gemini 3.6 Flash
- API Integration: @google/genai Node.js SDK via secure Express.js backend endpoints.

5. Docker Containerization & AWS Platform

- Containerization: Built using multi-stage Dockerfile (Node.js 20 Alpine).
- Cloud Platform: AWS Elastic Beanstalk (Docker on 64-bit Amazon Linux 2023).
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PART 2: PROJECT DEVELOPMENT REPORT

Vibe Coding Methodology & AWS Docker Deployment

1. Tech Stack Overview

- Frontend: React 19, Vite, Tailwind CSS v4, Lucide Icons, Motion.
- Backend: Node.js 20, Express.js, Esbuild, Mammoth parser.
- AI Engine: Google Gemini 3.6 Flash (@google/genai SDK).
- Containerization: Docker (Dockerfile based on Node.js 20 Alpine).
- Cloud: AWS Elastic Beanstalk (Docker on 64-bit Amazon Linux 2023).

2. Prompting Strategy & Frameworks

Persona Prompting

Configured AI with 'Expert Academic Exam Tutor' persona for clear, structured, encouraging responses.

JSON Schema Output

Enforced rigid JSON schema for Quiz Generation [{ id, question, options, correctIndex, explanation }].

3. Development Phases

Phase 1: Design & UI System

Implemented frosted-glass aesthetic with dark mode and mobile responsiveness.

Phase 2: Backend API Development

Built Express endpoints keeping Gemini API keys securely on server side.

Phase 3: Docker Containerization

Created Dockerfile packaging the full-stack app into a lightweight Docker container.

Phase 4: AWS Elastic Beanstalk Deployment

Launched AWS Elastic Beanstalk environment on Docker Platform with Environment Properties (PORT, NODE_ENV, GEMINI_API_KEY).

4. Reflections & Conclusion

Demonstrated end-to-end Vibe Coding methodology—architecting, containerizing with Docker, and deploying a functional AI web application on AWS Elastic Beanstalk Free Tier with zero cost and maximum security.